the time according to your computer time. It will set the present clock time of your system into the DS1307 I2C RTC module.

Method 2. *rtc.adjust(DateTime (2021, 7, 21, 10, 0, 0));* command sets the time into the RTC module in the format: year, month, date, hours, minutes, seconds (such as 2021, July 21, 10:00:00 am). Once the time is set, the backup battery in the RTC module keeps the time running for a long time.

To ring the bell and give the alert message at a particular time, say, at 9 hrs, 30 mins, the command should be as given below:

if (now.hour() = = 09 && now.minute() = = 30 && now.second() < = 15)

The 'if' statement is used for comparing with the set time for hours and minutes. The 'second() < = 15' is used to maintain the pulse high for a duration of 15 seconds to let the playback message to be completed. If you have a longer-duration playback message, change this time accordingly. In the 'if' loop, set the required timings for the alert message and the relay (buzzer) on and off timings.

After uploading the source code (schoolbell.ino) to Arduino Uno board, connect 12V DC power supply. The bell operates directly from 230V AC mains via the contacts of relay RL1 based on the scheduled programming time entered in the source code. You can feed multiple timings in the code as per the school-period timings.

PARTS LIST

Semiconductors:	
Board1	- Arduino Uno board
Miscellaneous:	
LS1	- 8-ohm,0.5-watt speaker
LS2	- 8-ohm,18-watt loudspeaker (optional)
Audio amplifier	- TDA2030A audio amplifier (optional)
MOD1	- 5V, single changeover (1C/O) relay
MOD2	- DS1307 RTC module
MOD3	- ISD1820 voice recorder/ playback module
	- 12V DC power supply
	- 230V AC school bell
	- Jumper wires
	- 2.1mm power jack

EFY Note
The source code of this project is available for free download at *source.efymag.com*

Every time the bell rings, after five seconds, a preprogrammed recorded massage (say, Alert, please maintain proper distance) of about ten seconds is announced via loudspeaker LS1. For a louder message, the optional audio amplifier, shown within dotted lines in the circuit, can be used along with loudspeaker LS2.

Arduino Uno board is highly sensitive, so handle it carefully. Check the DC power supply jack's input polarity (2.1mm centre-positive plug into the board's power jack) before use. EFY

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